

YUVARAJ S

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LinkedIn

GitHub

Portfolio

Professional Summary

AI/ML Engineer with specialized expertise in computer vision, natural language processing, and full-stack integration. Demonstrated success in developing production-ready machine learning solutions that solve real business challenges. Skilled in optimizing AI models for real-time applications with a focus on practical implementation and measurable performance metrics.

Skills

Technical

- TensorFlow, PyTorch, Scikit-learn
- Git & GitHub
- C, Python
- HTML, CSS
- UI Designing - Figma
- Machine Learning Algorithms
- OpenCV, YOLO
- BERT, NLP, Transformers
- Django, REST APIs
- AWS AI Foundations

Non-Technical

- Active Learner
- Innovation and Creativity
- Good Communication
- Leadership
- Problem Solving
- Logical Reasoning
- Team Collaboration
- Time Management
- Adaptability

Professional Experience

AI Machine Learning Engineer

July 2024 – October 2024

Techso IT LLC USA

- Engineered a Django-based real-time audio transcription system using SpeechRecognition, achieving 83%+ transcription accuracy on enterprise audio datasets
- Developed and deployed NLP-driven text summarization and multilingual translation modules (BART & mBART), reducing content review time by 40% and enabling seamless cross-language communication

Hackathons & Competitions

AI-Web Development Engineer

October 2024

Flipkart Robotic Challenge National Competition

- Designed and integrated 5 different AI models into unified platform with <100ms latency
- Developed RESTful API endpoints for seamless AI-frontend communication

Machine Learning Developer

September 2024

Amazon ML Challenge National Competition

- Developed NLP models to extract product attributes from unstructured text with 82% accuracy
- Implemented BERT-based feature extraction techniques for e-commerce product categorization

AI Solutions Developer

July 2023

IBM Hackathon Regional Competition

- Created facial emotion recognition system with real-time processing capabilities
- Built recommendation algorithms connecting detected emotions with personalized suggestions

Projects

Open to collaboration: send a GitHub collaborator request to view/work on projects: <https://github.com/Yuvaraj878>

- LetterTrack – One platform. Every letter. Total control.** | *Django & Workflow Automation*  [Repo](#)**Apr 2025 – Present**

- Built a centralized letter management platform to create, track, approve, and archive official letters end-to-end.
 - Implemented role-based access for Admin, Staff, and Students with controlled permissions and audit-friendly status tracking.
 - Designed workflow stages (Draft → Review → Approval → Issued) with notifications and history logs.
 - Added searchable records, filters, and export options to improve visibility and reduce manual follow-ups.
- Automated Library Access & Attendance System** | *Android & Django*  [Repo](#)**Nov 2024 – Apr 2025**

- Built a mobile check-in/checkout system using ID scan and face capture for secure access.
 - Created a central backend to record entries/exits and manage staff/students and departments.
 - Cut manual logging time by **85%** and gave teams real-time visibility into library usage.
 - Added easy CSV reports (daily, monthly, yearly, custom) and simple bulk data upload.
- FlipVision AI: E-Commerce Personalization** | *Computer Vision & ML*  [Repo](#)**Oct 2024 – Present**

- Designed and integrated 5 different AI models into a unified platform with <100ms latency.
 - Developed RESTful API endpoints for seamless AI-frontend communication.
- Appointment Booking Chatbot** | *Django & Conversational Flow*  [Repo](#)**Aug 2024 – Sep 2024**

- Developed a multi-step chatbot to guide patients through booking medical appointments seamlessly.
 - Integrated department and doctor selection, real-time slot availability checks, and symptom collection.
 - Implemented appointment scheduling with slot validation, automatic booking, and confirmation handling.
 - Improved scheduling efficiency and reduced manual staff workload by automating the entire booking flow.
- EchoFlow: Audio Analysis Platform** | *NLP & Signal Processing*  [Repo](#)**Jul 2024 – Aug 2024**

- Designed a multi-stage audio processing pipeline for real-time transcription and analysis.
 - Implemented BERT-based sentiment analysis on transcribed content with 87% accuracy.
- Mentor Selection System** | *Django & Web Development*  [Repo](#)**Feb 2023 – Oct 2023**

- Developed a web application for managing mentor-learner relationships with role-based access control.
 - Implemented Django backend with HTML, CSS, Bootstrap frontend supporting learner-mentor matching and HOD oversight.

Education

- B.Tech in Artificial Intelligence and Machine Learning****August 2021 – May 2026**

*Saveetha Engineering College**Chennai, India*

- Specialization: Deep Learning Architectures, Computer Vision, Natural Language Processing
 - Research Focus: Multi-modal Learning and Real-time AI Systems for Practical Applications
 - Key Coursework: Advanced Machine Learning, Neural Networks, Computer Vision, NLP, Data Structures
 - Active member of AI/ML community with 200+ members, participating in workshops and coding competitions

Professional Certifications & Achievements

- Efficient Large Language Model (LLM) Customization** — NVIDIA — May 2025
- AWS Certified AI Practitioner** — Amazon Web Services — February 2025
- Oracle Cloud Infrastructure 2024 AI Foundations Associate** — Oracle — January 2025
- Deep Learning Specialization** — Coursera — November 2023 — Including: CNNs, TensorFlow, NLP
- Secured top rankings** in national-level AI/ML competitions and hackathons